PAPER ID-310034

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VII) THEORY EXAMINATION 2024-25
DESIGN OF STEEL STRUCTURES

TIME: 3 HRS **M.MARKS: 100**

Note: Attempt all Sections. In case of any missing data; choose suitably.
IS 800 & Steel table allowed.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q no.	Question	CO	Level
a.	Write the mechanical properties of structural; steels	1	K1
b.	Where structural steel tubes are used?	1	K1
c.	Give any four suitable cases where fillet weld joints are used.	2	K1
d.	Write the classifications of bolted joints.	2	K1
e.	Give the list of different types of elements are used as tension members.	3	K1
f.	What are the rules to be following in the design of a tension member?	3	K1
g.	Distinguish column and strut.	4	K1
h.	Draw the diagram of buckling of column and mention its significance.	4	K1
i.	What is meant by Composite Beam Stage	5	K1
j.	What is the shear lag effect in beam flanges?	5	K1

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 30**

Q no.	Question	CO	Level
a.	Define Limit state & explain various types of limit state.	1	K1
b.	Calculate the bolt value of a 20mm diameter bolt of grade 4.6 to join main plates of 12mm thick under the following cases. Assume Fe410 for all plates. (i) Lap joint (ii) Single cover butt joint with cover plate = 10mm thickness	2	K2
c.	A double angle section back-to-back with @ Isa 90 x 90 x 10 is riveted to a 20 mmm gusset plate on one side. Determine the section capacity in tension and also the number of rivets required to develop the 85% tension capacity.	3	K2
d.	Write about batten plates compression member. Draw the column base plate diagram.	4	K3
e.	A simply supported beam of span 6 m is subjected to a concentrated load of 150 kN superimposed at mid span. Design the beam with rolled steel sections of f_y= 250 N/mm^2.	5	K2

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

Q no.	Question	CO	Level
a.	Assuming Fe410 steel, classify the following sections: (i) ISLB 350 @ 472 N/m (ii) ISHB 400 @ 806 N/m. Given that For Fe410, f_u=410 N/mm2, f_y= 250 N/mm^2.	1	K2

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b.	With neat sketch explain web buckling and web bearing failure.	1	K1

4. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	Explain application of different types of riveted joints. Also write the disadvantages of riveted joints.	2	K3
b.	A tie member consists of 2 ISMC 250. The channels are connected on either side of a 12 mm thick gusset plate. Design the welded joint to develop the full strength of the tie. However, the overlap is to be limited 400 mm.	2	K2

5. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	A tie of a roof truss consists of double angles ISA 100 X 75 X 10 mm with its short legs back-to-back and long legs connected to the same side of a gusset plate, with 16 mm diameter rivets. Determine the strength of tie in axial tension, taking ft = 150 N/mm^2. take rivets have been provided at suitable pitch.	3	K4
b.	How Angle sections eccentrically loaded through gussets plate?	3	K1

6. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	Design a column using two elements jointed together for a load of P= 100kN and length L = 8 m hinged at top and fixed at base. Use ISMB sections.	4	K2
b.	Explain the step-by-step procedure for finding the load carrying capacity of a compression member.	4	K2

7. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	A roof slab is supported by 6.25 m clear span beams spaced at 3.5 m apart. The thickness of RCC slab is 100 mm and the thickness of WPT of lime terracing with tiles is 90 mm. Only ISMB 250 with channel sections are available. Design the beam with f_y = 250 MPa.	5	K2
b.	With neat sketch explain the forces acting on gantry girders.	5	K2

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